4.5.2 The photoelectric effect

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** photoelectric effect, including a simple experiment to demonstrate this effect	Learners should understand that photoelectric effect provides evidence for particulate nature of electromagnetic radiation. HSW1, HSW2, HSW3, HSW7 Internet research on the development of quantum physics.
	(ii) demonstration of the photoelectric effect using, e.g. gold-leaf electroscope and zinc plate	
(b)	a one-to-one interaction between a photon and a surface electron	
(c)	Einstein's photoelectric equation	*M2.3*

(d) work function; threshold frequency

(e) the idea that the maximum kinetic energy of the photoelectrons is independent of the intensity of the incident radiation

(f) the idea that rate of emission of photoelectrons above the threshold frequency is directly proportional to the intensity of the incident radiation.